

BACKLOG BUDDY ? STUDY MATERIAL

Engineering Mechanics ? Unit-wise Lecture Notes

Subject Code: ME105 Semester: 2

.....
Condensed notes covering all five units: Statics of Particles, Equilibrium of Rigid Bodies, Centroid and Moment of Inertia, Dynamics of Particles, Friction and Virtual Work.

Unit 1: Statics of Particles

- Definition, key concepts and significance of Statics of Particles
- Important diagrams / derivations / algorithms for Statics of Particles
- Solved examples and previous-year questions on Statics of Particles

Unit 2: Equilibrium of Rigid Bodies

- Definition, key concepts and significance of Equilibrium of Rigid Bodies
- Important diagrams / derivations / algorithms for Equilibrium of Rigid Bodies
- Solved examples and previous-year questions on Equilibrium of Rigid Bodies

Unit 3: Centroid and Moment of Inertia

- Definition, key concepts and significance of Centroid and Moment of Inertia
- Important diagrams / derivations / algorithms for Centroid and Moment of Inertia
- Solved examples and previous-year questions on Centroid and Moment of Inertia

Unit 4: Dynamics of Particles

- Definition, key concepts and significance of Dynamics of Particles
- Important diagrams / derivations / algorithms for Dynamics of Particles
- Solved examples and previous-year questions on Dynamics of Particles

Unit 5: Friction and Virtual Work

- Definition, key concepts and significance of Friction and Virtual Work
- Important diagrams / derivations / algorithms for Friction and Virtual Work
- Solved examples and previous-year questions on Friction and Virtual Work

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.